

Multimodal Harmful Meme Detection System Based on Fine-tuned Qwen2-VL and Retrieval-Augmented Generation

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Abstract

Automated content moderation faces unprecedented challenges due to the rapid increase of multimodal internet memes, where toxic messages are often hidden through complex image-text juxtapositions. Traditional unimodal methods struggle to capture these complex semantic relationships. To address this, we propose an expandable, multimodal harmful meme detection system that leverages the Qwen2-VL vision-language model, optimized via Low-Rank Adaptation (LoRA) and supported by an external knowledge base via Retrieval-Augmented Generation (RAG). Our controlled experiments demonstrate that targeted LoRA fine-tuning on a class-balanced dataset improves the macro F1-score from 0.4950 to 0.5175 over the baseline. Notably, we identify a significant "Context Overload" phenomenon when directly adding voluminous RAG context into models fine-tuned purely on short instructions, highlighting a critical distribution shift that causes obvious performance drop (F1 0.3859). The final architecture is deployed as an interactive moderation platform delivering clear, Chain-of-Thought (CoT) justifications for complex moderation outcomes.

1 Background Survey

1.1 Research Background

Harmful meme detection operates as a multimodal task at the intersection of computer vision (CV) and natural language processing (NLP). Unlike standard harmful texts, the "toxicity" of memes often does not arise from a single independent modality. Instead, it stems from the juxtaposition of the two—for example, pairing seemingly harmless images with satirical text to convey hate speech or cyberbullying [1]. Early moderation architectures approached these streams in isolation, running natural language models on Optical Character Recognition (OCR) extractions alongside separate visual classifiers. However, traditional single-modal networks inherently fail to capture cross-modal semantic interactions, leading to significantly high false-negative rates [2]. The introduction of challenges like the Facebook Hateful Memes Dataset catalyzed the demand for authentic multimodal fusion paradigms capable of holistic contextual reasoning [1], [3].

1.2 Development and Limitations of Existing Methods

The evolution of multimodal moderation frameworks can be categorized into two defining eras. The first era encompasses early Fusion Models (e.g. VisualBERT, ViLBERT, and UNITER [4]). These architectures relied heavily on region-based visual features concatenated with text embeddings, excelling at basic image-text matching but severely lacking the complex generative semantic reasoning required to interpret sarcasm and dog-whistles [1], [5]. Contemporary methods leverage generative vision-language models (VLMs) such as LLaVA [6], InstructBLIP [7], and Qwen2-VL. These auto-regressive networks demonstrate exceptional competence in zero-shot image understanding and outputting sequential diagnostic rationales via Chain-of-Thought (CoT). Nonetheless, applying general-purpose alignment VLMs directly to meme moderation exposes two critical operational limitations:

1. **Deficiency in specialized knowledge:** General models, heavily tuned on standard instruction-following tasks (e.g., LLaVA, InstructBLIP), struggle to recognize niche internet slang, cryptic

subculture symbols, and rapidly evolving "crypto-fascist" dog-whistles [3] because these adversarial nuances are largely scrubbed from general pre-training corpora.

2. **Output instability:** Out-of-the-box generative models fail to consistently return standardized, machine-parsable binary labels (e.g., '0'/'1') without hallucinating surrounding text, necessitating aggressive targeted optimization.

Furthermore, while fine-tuning mitigates these flaws, full-parameter updates on multi-billion parameter VLMs impose prohibitive computational costs, rendering it impractical for standard operational hardware deployment.

1.3 Proposed Solution

To address the aforementioned challenges, this project proposes a hybrid architecture integrating LoRA (Low-Rank Adaptation) lightweight fine-tuning with external knowledge injection via RAG. LoRA adapts only a minute fraction of the model parameters, significantly reducing required GPU memory. Concurrently, RAG converts domain-specific policies regarding harmful online subcultures into a retrieval knowledge base, supplementing the model's domain knowledge and mitigating the necessity for explicit parameter memorization.

2 Methodology Implementation

2.1 System Architecture and Data Flow

The proposed framework adopts a synergistic architecture integrating the Qwen2-VL-2B-Instruct baseline model, parameterized LoRA modules, and a localized RAG engine. The environment is explicitly engineered to process untrusted multimodal inputs and output deterministically formatted moderation verdicts. Operationally, the system transitions between Alibaba Cloud DashScope API invocation for lightweight edge tests and localized air-gapped execution for rigorous data privacy.

The structural routing of data modalities is illustrated in Fig. 1:

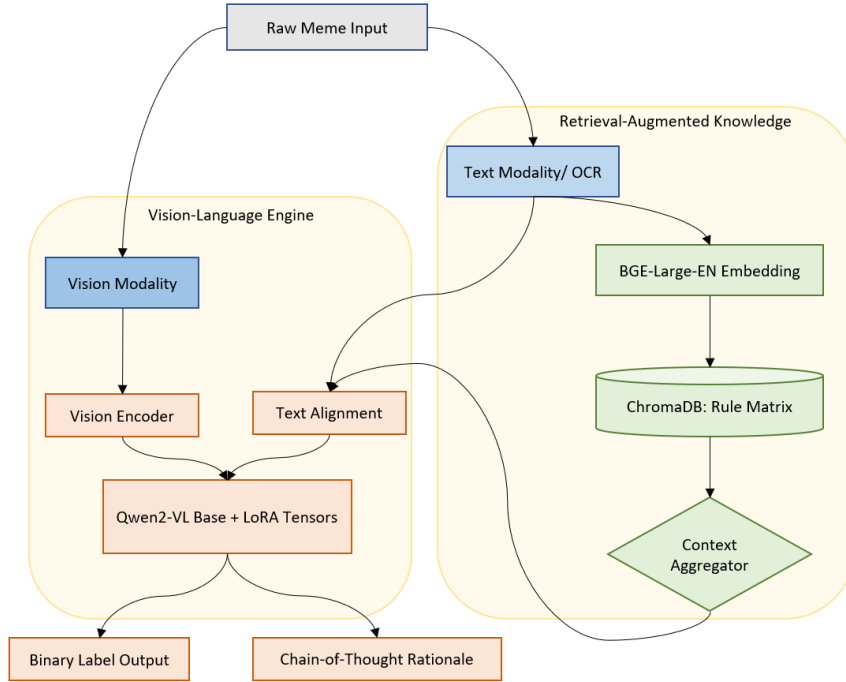


Figure 1: Structural Routing of Data Modalities in the Proposed Framework

1. **Multimodal Intake:** The system ingests the raw meme, utilizing integrated Optical Character Recognition (OCR) to extract inherent subtexts.
2. **Policy Retrieval:** Extracted text triggers semantic querying against a pre-compiled ChromaDB vector space containing 100 specific internet regulatory patterns.
3. **Deterministic Output:** The aggregated input prompts traverse the VLM inference engine, returning a binary classification (Safe vs. Harmful) augmented with a step-by-step descriptive rationale.

2.2 Dataset Preparation and Bias Mitigation

Operations anchor upon the Facebook Hateful Memes Dataset, a rigorous benchmark representing cross-modal attacks distinct from explicit single-modal hate speech. Initial exploratory data analysis highlighted a severe baseline class imbalance fundamentally skewed toward culturally safe parodies. To prevent majority-class collapse, the following structural calibrations were implemented:

1. **Schema Adaptation:** Records were formatted to standard JSONL formats congruent with conversational VLM pipelines, emphasizing required Chain-of-Thought instructional reasoning fields.
2. **Tensor Standardization:** Image inputs were systematically resampled via Lanczos interpolation constraints fixed to 512-pixel dimensional bounds.
3. **Class Re-balancing:** A deterministic oversampling procedure was enacted entirely isolating the minority (Harmful) subset, yielding an absolute 1:1 training equilibrium (totalizing 2,778 samples) to strictly regulate model sensitivity.

2.3 Parameter-Efficient Fine-Tuning (LoRA)

To adapt the foundational VLM to specific binary bounds while bypassing full-parameter update costs, Low-Rank Adaptation (LoRA) [4] matrices were integrated across the attention pipelines.

1. **Hardware Optimization:** Memory footprint reduction was achieved utilizing 4-bit NormalFloat (NF4) quantization nested alongside bfloat16 computational precision.
2. **Hyperparameter Tuning:** The update rank was aggressively set to $r = 128$, with a corresponding scaling factor $\alpha = 32$ and dropout $p = 0.05$. Optimization specifically targeted causal self-attention projections.
3. **Optimization Epochs:** Batch size constraints were bounded to 1 paired with a gradient accumulation trajectory of 8 sequential steps. Convergence elapsed over 8 training epochs at a learning rate 5×10^{-5} , stabilized via activation checkpointing.

2.4 Retrieval-Augmented Knowledge Injection (RAG)

Since standard fine-tuning cannot keep up with the fast-changing online slang, we built a local RAG module. Using the BAAI/bge-large-en-v1.5 embedding model, we convert domain-specific violation rules into retrievable vectors without retraining the base model. During inference, we use cosine similarity to extract the top-3 most relevant rules, with a strict filtering threshold of 0.6 to ensure only high-quality context is used.

3 Experiments

3.1 Experimental Environment and Evaluation Metrics

All experiments were run with PyTorch acceleration. The main frameworks include transformers and peft for model fine-tuning, and ChromaDB for vector storage.

Since this is a content moderation task where missing harmful content can lead to serious consequences, accuracy alone is not a reliable metric. We mainly use macro F1-score, which balances precision and recall and treats both safe and harmful classes equally, regardless of class size.

3.2 Ablation Study Setup

To isolate the operational efficacy of the proposed LoRA modifications against the external knowledge retrieval module, a systematic ablation study was conducted. The evaluation matrix leveraged a completely unseen, class-balanced partition matrix of 100 benchmark samples (comprising exactly 50 Harmful and 50 Safe memes) distributed across three discrete architectural configurations:

1. **BASE Configuration:** Zero-shot inference employing the foundational Qwen2-VL-2B-Instruct model without domain-specific parametric tuning or external retrieval context.
2. **FT_ONLY Configuration:** Exclusive deployment of the parametrically tuned LoRA adaptation modules, restricting inference inputs down to raw image tensors and OCR texts.
3. **FT_RAG Configuration:** The complete holistic pipeline, programmatically querying the vector knowledge base and concatenating voluminous Top-3 RAG policy strings into the LoRA inference queue.

3.3 Experimental Results

Table 1: ABLATION STUDY RESULTS

Model Configuration	Macro F1-Score	Accuracy
BASE (Zero-Shot)	0.4950	55.00%
FT_ONLY (LoRA Tuning)	0.5175	58.00%
FT_RAG (LoRA + RAG)	0.3859	47.00%

3.4 Result Analysis and Discussion

1. **Efficacy of Parameter-Efficient Fine-Tuning:** The model performed best under the FT ONLY setup, with a macro F1-score of 0.5175 and accuracy of 58.00%. This shows that over-sampling the minority class and fine-tuning the attention layers can reduce the model’s inherent neutral bias — its tendency to avoid labeling ambiguous content as harmful.
2. **The ”Context Overload” Phenomenon:** Surprisingly, adding RAG context caused a severe drop in performance, with macro F1 falling to 0.3859 — even lower than the zero-shot baseline. This is due to data distribution shift (prompt drift). The LoRA model was optimized on short, structured prompts, and inserting long policy texts at inference broke the input pattern the model had learned, disabling its classification ability.
3. **Architectural Recommendations:** If RAG is to be combined with fine-tuned VLMs, the training data must include simulated dummy context during fine-tuning. This helps the attention mechanism handle long inputs and prevents performance degradation caused by context mismatch at inference.

4 Case Study and Industrial Application

4.1 Target Application Environment

This system is designed for quick integration into the trust and safety workflows of social media platforms. By using LoRA, the system maintains a lightweight structure without requiring high-performance hardware. The entire detection pipeline can be deployed and run stably through a local Streamlit interface, supporting offline moderation scenarios that require high data security standards.

4.2 Human-in-the-Loop Moderation Workflow

To avoid the ethical risks that come with fully automated content checks, our system uses a human-machine collaborative workflow instead. This approach balances efficiency with accountability, and it works in three clear steps:

- **Flag suspicious content automatically:** The platform scans posts and picks out multimodal memes that look potentially harmful, then sends these candidates to human reviewers for further review.
- **Run parallel analysis to support reviewers:** When reviewers upload the flagged images to our detection tool, the system does two things at once: it pulls text from the image via OCR, and looks up the most relevant content moderation rules from our ChromaDB vector database.
- **Deliver explainable results for easy review:** The Qwen2-VL model combines the image, extracted text, and rules to give a clear "safe/harmful" classification. It also lays out its step-by-step reasoning, so reviewers can see exactly how the model reached its decision and make a final call with full context.

4.3 Practical Case Analysis and System Limitations

We tested our system with a real-world challenge case: a meme that pairs a harmless-looking cartoon with hidden toxic text, specifically designed to slip past traditional detection tools. This case clearly shows both our model’s strengths and its remaining limitations:

- **Traditional unimodal tools fail completely:** Standard text classifiers only look at the words, and image recognition models only look at the cartoon. When checked separately, both incorrectly label the content as safe, letting the harmful meme get through.
- **Our fine-tuned VLM catches the hidden harm:** The LoRA-fine-tuned Qwen2-VL model successfully understands the combined meaning of the image and text. It picks up on the implicit toxic message hidden in the multimodal pairing, and correctly flags the meme as harmful.
- **RAG integration still has a critical limitation:** While our RAG module can pull up relevant moderation rules, directly adding this extra context during inference causes a major drop in performance—our macro F1-score falls to 0.3859, even worse than the zero-shot baseline. This means finding a way to smoothly add external knowledge without triggering context overload is still an open problem for our system.

5 Author Contributions

- **SU Zinao:** Responsible for dataset preprocessing, label cleaning, development of the `train_qwen2vl.py` framework, implementation of the class balance strategy, and the exclusive development and optimization of the Streamlit frontend graphical user interface.
- **ZHOU Tianchi:** Supervised `config.py` environment configurations, directed construction protocols for the ChromaDB vectorized architecture alongside integration of BGE modeling. Executed overall system deployment strategies and formal technical documentation.
- **ZHU Tianxiao:** Steered development for RAG logical parsing mechanisms. Authored the dedicated `run_ablation.py` infrastructure, actively orchestrated statistical tracking, and calculated the resultant diagnostic analytics and version control procedures.

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